

Pattern Program

Steps to Solve Pattern.

1) Identify the Shape → No of rows & No of Column.

2) Characters

3) Sequence

*	*	*	*

```
public class Pattern1 {  
    public static void main(String[] args) {  
        int n=4;  
        for (int i = 0; i <n; i++) {  
            System.out.print("* ");  
        }  
    }  
}
```

	0	1	2	3
0	*			
1	*			
2	*			
3	*			

```

public class Pattern2 {
public static void main(String[] args) {
    int n=4;
    for (int i = 0; i <n; i++) {

        System.out.println("* ");

    }
}
}

```

	0	1	2	3
0	*	*	*	*
1	*	*	*	*
2	*	*	*	*
3	*	*	*	*

```

public class Pattern3 {
public static void main(String[] args) {
    int n=4;
    for (int i = 0; i <n; i++) {

        for (int j = 0; j < n; j++) {
            System.out.print("* ");
        }
        System.out.println();
    }
}
}

```

	0	1	2	3
0	*			
1	*	*		
2	*	*	*	
3	*	*	*	*

```

public class Pattern1 {
public static void main(String[] args) {
    int n=4;
    for (int i = 0; i <n; i++) {

        for (int j = 0; j < n; j++) {
            if (i>=j) {
                System.out.print("* ");
            }
        }
        System.out.println();
    }
}
}

```

	0	1	2	3
0	*	*	*	*
1	*	*	*	
2	*	*		
3	*			

```

public class Pattern1 {
public static void main(String[] args) {
    int n=4;
    for (int i = 0; i <n; i++) {

        for (int j = 0; j < n; j++) {
            if (i<=j) {
                System.out.print("* ");
            }
        }
        System.out.println();
    }
}
}

```

	0	1	2	3
0	*	*	*	*
1		*	*	*
2			*	*
3				*

```

public class P8 {
    public static void main(String[] args) {
        int n=4;
        for (int i = 0; i < 4; i++) {
            for (int j = 0; j < 4; j++) {
                if (i<=j) {
                    System.out.print("*");
                } else {
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
    }
}

```

	0	1	2	3
0	*			
1	*	*		
2	*	*	*	
3	*	*	*	*

```

public class P7 {
    public static void main(String[] args) {
        int n=4;
        for (int i = 0; i < 4; i++) {
            for (int j = 0; j < 4; j++) {
                if (i>=j) {
                    System.out.print("*");
                } else {
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
    }
}

```

WAP to Print the Following Pattern

	0	1	2	3
0				*
1			*	
2		*		
3	*			

```
public class P8 {  
    public static void main(String[] args) {  
        int n=4;  
        for (int i = 0; i < 4; i++) {  
            for (int j = 0; j < 4; j++) {  
                if (i==j) {  
                    System.out.print("*");  
                } else {  
                    System.out.print(" ");  
                }  
            }  
            System.out.println();  
        }  
    }  
}
```


1)WAIJP to Print the following pattern

	0	1	2	3
0	*			
1		*		
2			*	
3				*

```
public class P8 {  
    public static void main(String[] args) {  
        int n=4;  
        for (int i = 0; i < 4; i++) {  
            for (int j = 0; j < 4; j++) {  
                if (i+j==n-1) {  
                    System.out.print("*");  
                } else {  
                    System.out.print(" ");  
                }  
            }  
            System.out.println();  
        }  
    }  
}
```

2)WAIJP to Print the following pattern

	0	1	2	3
0	*	*	*	*
1	*			*
2	*			*
3	*	*	*	*

```
public class P9 {  
    public static void main(String[] args) {  
        int n=4;  
        for (int i = 0; i < 4; i++) {  
            for (int j = 0; j < 4; j++) {  
                if (i==0||i==n-1||j==0||j==n-1)  
{  
                    System.out.print("*");  
                } else {  
                    System.out.print(" ");  
                }  
            }  
            System.out.println();  
        }  
    }  
}
```

3)WBJP to Print the following pattern

```
* * * * *
* *           * *
*   *       *   *
*       *   *       *
*           *       *
*       *   *       *
*   *       *   *
* *           * *
* * * * *
```

```
public class P10 {
public static void main(String[] args) {
    int n=9;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (i==0||j==0||i==n-1||j==n-1||i==j
||i+j==n-1) {
                System.out.print("* ");
            } else {
                System.out.print("  ");
            }
        }
        System.out.println();
    }
}
```

4)WBJP to Print the Following Pattern

```
* * * * * * * * * * *
*           *           *
*           *           *
*           *           *
*           *           *
* * * * * * * * * * *
*           *           *
*           *           *
*           *           *
*           *           *
* * * * * * * * * * *
```

```
public class P9 {
    public static void main(String[] args) {
        int n=11;
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (i==0||j==0||i==n-1||j==n-1||i==n/2||j==n/2) {
                    System.out.print("* ");
                } else {
                    System.out.print("  ");
                }
            }
            System.out.println();
        }
    }
}
```

5)WAP to print the following pattern

```
* * * * *
* *      *      * *
*  *      *      *
*      *  *  *      *
*      * * *      *
* * * * *
*      * * *      *
*      *  *      *
*      *      *      *
* *      *      * *
* * * * *
* * * * *
```

```
public class P10 {
    public static void main(String[] args) {
        int n=11;
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (i==0||j==0||i==n-1||j==n-1||i==n/2||j==n/2||i==j||i+j==n-1) {
                    System.out.print("* ");
                } else {
                    System.out.print("  ");
                }
            }
            System.out.println();
        }
    }
}
```

6)WAP to Print the following pattern

```
*           *
* *         * *
* * *       * * *
* * * * *   * * * * *
```

```
public class P1 {
public static void main(String[] args) {
    int n=4;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (i>=j) {
                System.out.print("* ");
            } else {
                System.out.print("  ");
            }
        }
        for (int j = 0; j < n; j++) {
            if (i>=j) {
                System.out.print("* ");
            }
        }
        System.out.println();
    }
}
}
```

7)WAP to print the following pattern

```
*
* *
* * *
* * * * *
```

```
public class P1 {
    public static void main(String[] args) {
        int n=4;
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (i>=j) {
                    System.out.print("* ");
                } else {
                    System.out.print(" ");
                }
            }
            for (int j = 0; j < n; j++) {
                if (i+j>=n-1) {
                    System.out.print("* ");
                }
                else
                {
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
    }
}
```

8)WAP to Print the Following Pattern

```
*           * * * *
* *         * * *
* * *       * *
* * * *     *
```

```
public class P1 {
public static void main(String[] args) {
    int n=4;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (i>=j) {
                System.out.print("* ");
            } else {
                System.out.print("  ");
            }
        }
        for (int j = 0; j < n; j++) {
            if (i<=j) {
                System.out.print("* ");
            }
            else
            {
                System.out.print("  ");
            }
        }
        System.out.println();
    }
}
}
```


9)WAP to Print the following pattern

```
*           * * * *
* *         * * *
* * *       * *
* * * * *   *
```

```
public class P1 {
public static void main(String[] args) {
    int n=4;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (i>=j) {
                System.out.print("* ");
            } else {
                System.out.print(" ");
            }
        }
        for (int j = 0; j < n; j++) {
            if (i+j<=n-1) {
                System.out.print("* ");
            }
            else
            {
                System.out.print(" ");
            }
        }
        System.out.println();
    }
}
}
```

10)WAP to Print the following pattern

```
*           *
* *         * *
* * *       * * *
* * * * *   * * * * *
```

```
public class P1 {
public static void main(String[] args) {
    int n=4;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (i>=j) {
                System.out.print("* ");
            } else {
                System.out.print(" ");
            }
        }
        for (int j = 0; j < n; j++) {
            if (i>=j) {
                System.out.print("* ");
            }
            else
            {
                System.out.print(" ");
            }
        }
        System.out.println();
    }
}
}
```

11)WAP to Print the following pattern

```
* * * * *
* * *   * * *
* *     * *
*       *
```

```
public class P1 {
public static void main(String[] args) {
    int n=4;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (i+j<=n-1) {
                System.out.print("* ");
            } else {
                System.out.print(" ");
            }
        }
        for (int j = 0; j < n; j++) {
            if (i<=j) {
                System.out.print("* ");
            }
            else
            {
                System.out.print(" ");
            }
        }
        System.out.println();
    }
}
}
```

12)WAP to print the following pattern

```
    * * * * *
  * *   * * *
* * *     * *
* * * *   *
```

```
public class P1 {
public static void main(String[] args) {
    int n=4;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (i+j>=n-1) {
                System.out.print("* ");
            } else {
                System.out.print("  ");
            }
        }
        for (int j = 0; j < n; j++) {
            if (i<=j) {
                System.out.print("* ");
            }
            else
            {
                System.out.print("  ");
            }
        }
        System.out.println();
    }
}
}
```

13)WAP to Print the following pattern

```
* * * *      *
  * * *      * *
    * *      * * *
      * * * * *
```

```
public class P1 {
public static void main(String[] args) {
    int n=4;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (i<=j) {
                System.out.print("* ");
            } else {
                System.out.print("  ");
            }
        }
        for (int j = 0; j < n; j++) {
            if (i+j>=n-1) {
                System.out.print("* ");
            }
            else
            {
                System.out.print("  ");
            }
        }
        System.out.println();
    }
}
}
```

14)WAP to Print the following pattern

```
* * * * *
 * * * * *
  * * * * *
   * * * * *
```

```
public class P1 {
public static void main(String[] args) {
    int n=4;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (i<=j) {
                System.out.print("* ");
            } else {
                System.out.print(" ");
            }
        }
        for (int j = 0; j < n; j++) {
            if (i>=j) {
                System.out.print("* ");
            }
            else
            {
                System.out.print(" ");
            }
        }
        System.out.println();
    }
}
}
```

15)WAP to Print the following pattern

```
    * * * * *
  * * * * *
* * * * *
* * * * *
```

```
public class P1 {
public static void main(String[] args) {
    int n=4;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (i+j>=n-1) {
                System.out.print("* ");
            } else {
                System.out.print("  ");
            }
        }
        for (int j = 0; j < n; j++) {
            if (i+j<=n-1) {
                System.out.print("* ");
            }
            else
            {
                System.out.print("  ");
            }
        }
        System.out.println();
    }
}
}
```

16)WAP to Print the following pattern

```
1 2 3 4
5 6 7 8
9 10 11 12
13 14 15 16
```

```
public class P2 {
    public static void main(String[] args) {
        int n=4;
        int a=1;
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                System.out.print(a++ + " ");
            }
            System.out.println();
        }
    }
}
```


17)WAP to Print the following pattern

```
A B C D
E F G H
I J K L
M N O P
```

```
public class P2 {
    public static void main(String[] args) {
        int n=4;
        char ch='A';
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                System.out.print(ch++ + " ");
            }
            System.out.println();
        }
    }
}
```

18)WAP to print the following pattern

```
1 2 3 4
A B C D
5 6 7 8
E F G H
```

```
public class P4 {
    public static void main(String[] args)
    {
        int n=4;
        char ch='A';
        int a=1;
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (i%2==0) {
                    System.out.print(a++ + " ");
                } else {
                    System.out.print(ch++ + " ");
                }
            }
            System.out.println();
        }
    }
}
```

19)WAIJP to print the following pattern

```
1 A 2 B
3 C 4 D
5 E 6 F
7 G 8 H
```

```
public class P4 {
    public static void main(String[] args)
    {
        int n=4;
        char ch='A';
        int a=1;
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (j%2==0) {
                    System.out.print(a++ + "");
                } else {
                    System.out.print(ch++ + "");
                }
            }
            System.out.println();
        }
    }
}
```

20)WAP to print the following pattern

```
1 2 3 4
1 2 3 4
1 2 3 4
1 2 3 4
```

```
public class P2 {
    public static void main(String[] args) {
        int n=4;

        for (int i = 0; i < n; i++) {
            int a=1;
            for (int j = 0; j < n; j++) {
                System.out.print(a++ + " ");
            }
            System.out.println();
        }
    }
}
```

21)WAP to print the following pattern

```
1 1 1 1
2 2 2 2
3 3 3 3
4 4 4 4
```

```
public class P2 {
    public static void main(String[] args) {
        int n=4;
        int a=1;
        for (int i = 0; i < n; i++) {

            for (int j = 0; j < n; j++) {
                System.out.print(a + " ");
            }
            a++;
            System.out.println();
        }
    }
}
```

22)WAP to print the following pattern

```
A B C D
A B C D
A B C D
A B C D
```

```
public class P2 {
    public static void main(String[] args) {
        int n=4;

        for (int i = 0; i < n; i++) {
            char ch='A';
            for (int j = 0; j < n; j++) {
                System.out.print(ch++ + " ");
            }

            System.out.println();
        }
    }
}
```

23)WAP to print the following pattern

```
A A A A
B B B B
C C C C
D D D D
```

```
public class P2 {
    public static void main(String[] args) {
        int n=4;
        char ch='A';
        for (int i = 0; i < n; i++) {

            for (int j = 0; j < n; j++) {
                System.out.print(ch + " ");
            }
            ch++;
            System.out.println();
        }
    }
}
```

24)WAP to print the following pattern

```
1 a 1 a 1
2 b 2 b 2
3 c 3 c 3
4 d 4 d 4
5 e 5 e 5
```

```
public class P5 {
public static void main(String[] args) {
    int n=5;
    int a=1;
    char ch='a';
    for (int i = 0; i <n; i++) {

        for (int j = 0; j < n; j++) {
            if (j%2==0) {
                System.out.print(a + " ");
            } else {
                System.out.print(ch + " ");
            }

        }

        a++;
        ch++;

        System.out.println();
    }
}
```


25)WAP to print the following pattern

```
1 2 3 4 5
a b c d e
$ $ $ $ $
a b c d e
1 2 3 4 5
```

```
public class P6 {
public static void main(String[] args) {
    int n=5;
    for (int i = 0; i < n; i++) {
        int a=1;
        char ch='a';
        for (int j = 0; j < n; j++) {
            if (i==2) {
                System.out.print("$" + " ");
            } else if(i%2==0){
                System.out.print(a++ + " ");
            }
            else
            {
                System.out.print(ch++ + " ");
            }
        }
        System.out.println();
    }
}
}
```

26)WAP to print the following pattern

```
4 3 2 1
4 3 2 1
4 3 2 1
4 3 2 1
```

```
public class P7 {
    public static void main(String[] args) {
        int n=4;

        for (int i = 0; i < n; i++) {
            int a=4;
            for (int j = 0; j < n; j++) {
                System.out.print(a-- + " ");
            }

            System.out.println();
        }
    }
}
```

27)WAP to print the following pattern

```
e d c b a
e d c b a
e d c b a
e d c b a
e d c b a
```

```
public class P8 {
    public static void main(String[] args) {
        int n=5;
        for (int i = 0; i < n; i++) {
            char ch='e';
            for (int j = 0; j < n; j++) {
                System.out.print(ch--+" ");
            }
            System.out.println();
        }
    }
}
```

28)WAP to print the following pattern

```
1
2 3
4 5 6
7 8 9 10
```

```
public class P9 {
public static void main(String[] args) {
    int n=4;
    int a=1;
    for (int i = 0; i < n; i++) {
        for (int j = 0; j < n; j++) {
            if (i>=j) {
                System.out.print(a++ + " ");
            }
        }
        System.out.println();
    }
}
}
```

29)WAP to print the following pattern

```
1
a b
2 3 4
c d e f
```

```
public class P10 {
    public static void main(String[] args) {
        int n=4;
        int a=1;
        char ch='a';
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (i>=j) {
                    if (i%2==0) {
                        System.out.print(a++ + " ");
                    } else {
                        System.out.print(ch++ + " ");
                    }
                }
            }
            System.out.println();
        }
    }
}
```

30)WAIJP to print the following pattern

```
1
a b
1 2 3
a b c d
```

```
public class P10 {
    public static void main(String[] args) {
        int n=4;

        for (int i = 0; i < n; i++) {
            int a=1;
            char ch='a';
            for (int j = 0; j < n; j++) {
                if (i>=j) {
                    if (i%2==0) {
                        System.out.print(a++ + " ");
                    } else {
                        System.out.print(ch++ + " ");
                    }
                }
            }
            System.out.println();
        }
    }
}
```

31)WAP to print the following pattern

```
    1
   2 3 4
  5 6 7 8 9
10 11 12 13 14 15 16
17 18 19 20 21 22 23 24 25
```

```
public class P15{
public static void main(String[] args) {
    triangle(5);
}
public static void triangle(int n)
{
    int a=1;
    for (int i = 1; i <=n; i++) {
        for (int j = 1; j <=n-i; j++) {
            System.out.print(" ");
        }
        for (int j = 1; j <= 2*i-1; j++) {
            System.out.print(a++ + " ");
        }
        System.out.println();
    }
}
}
```

32)WAP to print the following pattern

```
  *
 ***
*****
*****
*****
*****
```

```
public class P14 {
public static void main(String[] args) {
    triangle(6);
}
public static void triangle(int n)
{
    for (int i = 1; i <=n; i++) {
        for (int j = 1; j <=n-i; j++) {
            System.out.print(" ");
        }
        for (int j = 1; j <= 2*i-1; j++) {
            System.out.print("*");
        }
        System.out.println();
    }
}
}
```


33)WAP to print the following pattern

```
  A
 B C D
E F G H I
J K L M N O P
Q R S T U V W X Y
```

```
public class P16{
public static void main(String[] args) {
    triangle(5);
}
public static void triangle(int n)
{
    char ch='A';
    for (int i = 1; i <=n; i++) {
        for (int j = 1; j <=n-i; j++) {
            System.out.print(" ");
        }
        for (int j = 1; j <= 2*i-1; j++) {
            System.out.print(ch++ + " ");
        }
        System.out.println();
    }
}
}
```

34)WAP to print the following pattern

```
    1
  1 2 3
1 2 3 4 5
1 2 3 4 5 6 7
1 2 3 4 5 6 7 8 9
```

```
public class P10 {
    public static void main(String[] args) {
        int n=4;

        for (int i = 0; i < n; i++) {
            int a=1;
            char ch='a';
            for (int j = 0; j < n; j++) {
                if (i>=j) {
                    if (i%2==0) {
                        System.out.print(a++ + " ");
                    } else {
                        System.out.print(ch++ + " ");
                    }
                }
            }
            System.out.println();
        }
    }
}
```

35)WAP to print the following pattern

```
  1
 2 2 2
3 3 3 3 3
4 4 4 4 4 4 4
5 5 5 5 5 5 5 5 5
```

```
public class P18{
public static void main(String[] args) {
    triangle(5);
}
public static void triangle(int n)
{
    int a=1;
    for (int i = 1; i <=n; i++) {

        for (int j = 1; j <=n-i; j++) {
            System.out.print(" ");
        }
        for (int j = 1; j <= 2*i-1; j++) {
            System.out.print(a + " ");
        }
        a++;
        System.out.println();
    }

}
}
```

36)WAP to print the following pattern

```
1
101
10101
1010101
101010101
```

```
public class P19{
public static void main(String[] args) {
    triangle(5);
}
public static void triangle(int n)
{
    int a=1;
    for (int i = 1; i <=n; i++) {

        for (int j = 1; j <=n-i; j++) {
            System.out.print(" ");
        }
        for (int j = 1; j <= 2*i-1; j++) {
            if (j%2==1) {
                System.out.print("1");
            } else {
                System.out.print("0");
            }
        }

    }

    System.out.println();
}
}
```

37)WAP to print the following pattern

```
0
111
00000
1111111
000000000
```

```
public static void main(String[] args) {
    triangle(5);
}
public static void triangle(int n)
{
    int a=1;
    for (int i = 1; i <=n; i++) {

        for (int j = 1; j <=n-i; j++) {
            System.out.print(" ");
        }
        for (int j = 1; j <= 2*i-1; j++) {
            if (i%2==0) {
                System.out.print("1");
            } else {
                System.out.print("0");
            }
        }

    }

    System.out.println();
}

}
```

38)WAP to print the following pattern

```
    1
  1 2 1
1 2 3 2 1
1 2 3 4 3 2 1
1 2 3 4 5 4 3 2 1
```

```
public class P12 {
public static void main(String[] args) {
    int n=5;
    for (int i = 1; i <= n; i++) {
        for (int j = 1; j < n-i; j++) {
            System.out.print(" ");
        }
        int x=1;
        for (int j = 1; j <= 2*i-1 ; j++) {
            System.out.print(x+" ");
            if (j<i) {
                x++;
            } else {
                x--;
            }
        }
        System.out.println();
    }
}
}
```

```
public class P28 {
    public static void main(String[] args) {

        int n=10;
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (i+j==n-1||i==n-1) {
                    System.out.print("* ");
                } else {
                    System.out.print("  ");
                }
            }

            for (int j = 1; j < n ; j++) {

                if (i==j||i==n-1) {
                    System.out.print("* ");
                } else {
```

```

        System.out.print(" ");
    }
}
System.out.println();

}
}
}

```

39)WAP to print the following pattern

```

    1
  2 3 2
 3 4 5 4 3
4 5 6 7 6 5 4
5 6 7 8 9 8 7 6 5

```

```

public class P27 {
    public static void main(String[] args) {

        int n=5;
        for (int i = 1; i <= n; i++) {
            for (int j = 1; j < n-i; j++) {
                System.out.print(" ");
            }
            int x=i;
            for (int j = 1; j <= 2*i-1 ; j++) {
                System.out.print(x+" ");
                if (j<i) {

```



```

        x++;
    } else {
        x--;
    }
}
System.out.println();

}
}

}

```

40)WAP to Print the following pattern

```

A B C D
E       F
G       H
I J K L

```

```

public class P29 {
    public static void main(String[] args) {
        int n=4;
        char ch='A';
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (i==0 || i==n-1 || j==0 || j==n-1) {

```

```

        System.out.print(ch+" ");
        ch++;
    } else {
        System.out.print(" ");
    }
}
System.out.println();
}
}
}

```

41)WAP to print the following pattern

```

1 2 3 4
5      6
7      8
9 10 11 12

```

```

public class P29 {
    public static void main(String[] args) {
        int n=4;
        int a=1;
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (i==0||i==n-1||j==0||j==n-1) {
                    System.out.print(a+" ");
                    a++;
                } else {
                    System.out.print(" ");
                }
            }
            System.out.println();
        }
    }
}

```

```
public class P28 {
    public static void main(String[] args) {

        int n=10;
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (i==j||i==0) {
                    System.out.print("* ");
                } else {
                    System.out.print("  ");
                }
            }
        }
    }
}
```

```

    }
}

for (int j = 1; j < n ; j++) {
    if (i+j==n-1||i==0) {
        System.out.print("* ");
    } else {
        System.out.print("  ");
    }
}
System.out.println();

}
}
}

```

42)WAIJP to print the following pattern

```

      *
     * *
    * * *
   * * * *
  * * * * *
 * * * * *
* * * * *
 * * * * *
  * * * * *
   * * * *
    * * *
     * *
      *

```

```

public class P28 {
    public static void main(String[] args) {

        int n=10;
        for (int i = 0; i < n; i++) {
            for (int j = 0; j < n; j++) {
                if (i+j==n-1) {
                    System.out.print("* ");
                } else {

```

```

        System.out.print("  ");
    }

}

for (int j = 1; j < n ; j++) {

    if (i==j) {
        System.out.print("* ");
    } else {
        System.out.print("  ");
    }
}
System.out.println();

}

for (int i = 0; i < n; i++) {
    for (int j = 0; j < n; j++) {
        if (i==j) {
            System.out.print("* ");
        } else {
            System.out.print("  ");
        }
    }

}

for (int j = 1; j < n ; j++) {

    if (i+j==n-1) {
        System.out.print("* ");
    } else {
        System.out.print("  ");
    }
}

```

```

        }
        System.out.println();
    }
}
}

```

43)WAP to Print the following pattern

```

7 6 5 4 3 2 1
8 9 10 11 12 13
18 17 16 15 14
19 20 21 22
25 24 23
26 27
28

```

```

public class SnakePattern {
    public static void main(String[] args) {
        int n=8;
        int a=1;
        for (int i = 1; i < n; i++) {

```

```

        for (int j = 1; j < n; j++) {
            if (i<=j) {
                if (i%2==0) {
                    System.out.print(a++ + "
");
                }
                else {
                    a++;
                }
            }
        }
        int b=a-1;
        for (int j = 0; j <n; j++) {
            if (i<=j) {
                if (i%2==1) {
                    System.out.print(b-- + "
");
                }
            }
        }
        System.out.println();
    }
}
}

```